Agentic MT: Initial Experiments

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Six phases testing the two-pillar theory (slow thinking + tool use, jointly necessary, most valuable for low-resource pairs). Code and full per-phase logs: `github.com/imthomp/agentic-mt`.

Where the theory stands

The central joint-necessity claim has not been confirmed or falsified — every direct test has hit a model confound first. We do have real evidence against the simplest instantiations of two supporting mechanisms (QE-in-the-loop, source-side triggering), and a new finding, from one model across four language points (BLOOMZ, Phase 6), that raw pretraining exposure to a language can be a hard gate on generation, separate from anything the agentic scaffolding does.

Phase-by-phase

Phase	Setup	Finding
1: QE reliability	COMET-KIWI vs. real WMT DA human scores, 5 low-resource pairs (ha/km/ps/xh/zu), 1000-sample bootstrap	Below 0.3 for 3/5 pairs (ha 0.115, ps 0.295, zu 0.247); km/xh weak but usable
2: Triggering	5 source-side features + logistic regression on Phase 1's data, same 5 pairs ($n = 38786$)	No feature or model beats chance (AUC 0.513–0.564); pooled OOV signal ($r = -0.33$) is a Simpson's-paradox artifact
3: Joint necessity	Aya-101, 4 conditions $\times$ en-de/en-ha $\times$ 100 sentences	Tool-alone wins both pairs; CoT hurts; gap differs by resource level ($p = 0.004$–0.024) — but CoT chain degenerates (see Fig. 1)
4: Refinement	Qwen2.5-32B, TEaR self-critique vs. external-QE refine, en-de, 100 sentences, 4 iterations	Neither condition peaks-then-declines; both improve gradually to iteration 4, matching TEaR's actual reported pattern
5: Model search	Tried BLOOMZ, InkubaLM as open, conversation-tuned models covering ha/km/ps/xh/zu	Both ruled out: InkubaLM incoherent even in English (0.4 B params); BLOOMZ needed 9 fixes, then failed on exposure grounds
6: Exposure floor	BLOOMZ smoke test, en-fr/en-sw/en-xh/en-zu, 10 sentences/pair, hand-categorized	Clean-generation rate tracks ROOTS pretraining bytes monotonically, this model only (see Fig. 2)

Findings, by implication

Challenges to specific proposed mechanisms

- **QE-in-the-loop (Sec. 5.3) is not viable as specified for most of the 5 low-resource pairs tested (ha/km/ps/xh/zu).** An en-de benchmark of ≈ 0.57 sets the bar these pairs miss. To trust more: extend beyond the five pairs with real human judgments; current coverage is a hard ceiling on what we can check.
- **Source-side triggering (Sec. 5.1) shows no signal from cheap features on these same 5 pairs,** against a chance baseline of 0.5. The pooled OOV correlation evaporates to -0.045–0.021 once controlled for language. To trust more: a real domain classifier and a frequency-list OOV measure before calling triggering a dead end rather than an implementation gap.

The central joint-necessity claim: confounded, not resolved

- **Phase 3**: Aya-101 is not conversation-tuned, and its CoT chain produced byte-identical draft/refine/proofread output in 80–86 of 100 sentences (Fig. 1). The "CoT hurts" result is real in the sense that it's what happened, but is confounded with architecture, not a clean test of reasoning per se.
- **Phase 4**: matches Feng et al.'s actual TEaR pattern, not the early-peak hypothesis we set out to test (that pattern belongs to a different paper). Secondary finding: condition A froze only 31/100 sentences by iteration 4 vs. condition B's 48/100 by iteration 3, and A's trajectory is smoother — span-level feedback may matter more than feedback source. To trust more: rerun condition B with XCOMET (span-level) to isolate granularity from source.
- **Phases 5–6**: two replacement-model candidates ruled out with direct evidence, not assumption (see table). **Net: the joint-necessity question is still untested on a model without a disqualifying confound.**

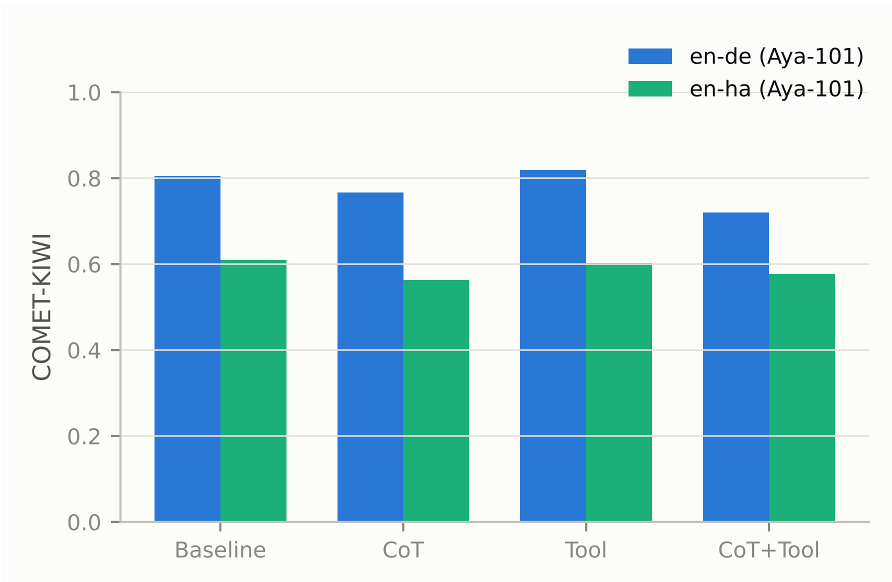

Figure 1: Phase 3 condition means (COMET-KIWI). Tool wins both pairs; CoT drops both — but see the degeneracy caveat above.

Tangential but new: a pretraining-exposure floor

- **Phase 6**: below some exposure level BLOOMZ doesn't attempt the target language at all — it defaults to echoing the English source (comprehension intact, generation not) — rather than degrading gracefully. To trust more: only 4 points on the curve; need intermediate-exposure languages (e.g. Yoruba, ~90M bytes) to call it an actual regression rather than an eyeballed trend.
- Cross-checked for Phase 3's Aya-101/mT5: German has 347B tokens in mC4 vs. Hausa's 0.2B ($\sim 1735\times$) — so Phase 3's en-de anchor *is* legitimate for that model, unlike BLOOM's zero-exposure German. Same question, opposite answer for the two models — check per-model, don't assume.

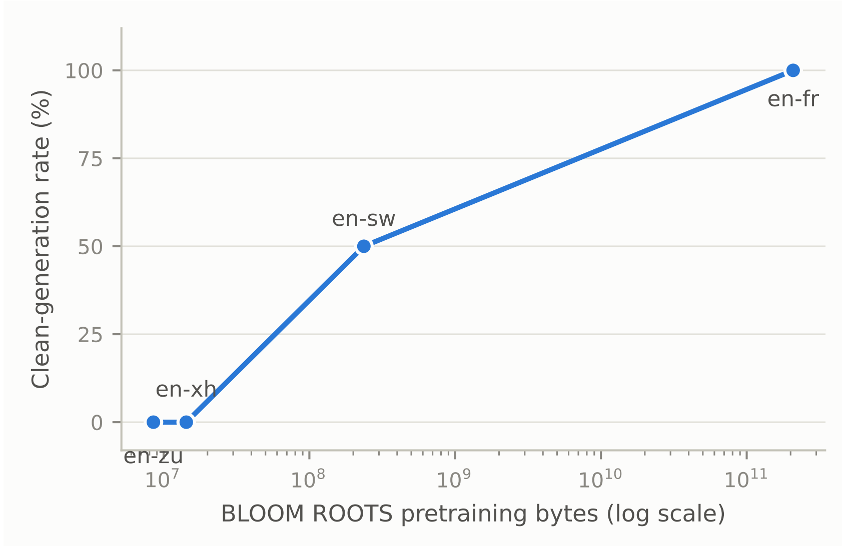

Figure 2: BLOOMZ clean-generation rate vs. ROOTS pretraining bytes (log scale), en-fr/en-sw/en-xh/en-zu.

Open decision

Three ways to get an unconfounded joint-necessity result:

1. **Keep searching** for an open model that is conversation-tuned and covers the target languages. Two candidates already ruled out; no known third, and preserves an all-open-source stack.
2. **Split the two roles**: an API model for reasoning, a Church-fine-tuned NLLB model queried as the tool. Sidesteps Phase 6's exposure floor (NLLB-200 targets all five pairs by design, though we have not verified its per-language training scale the way Phase 6 did for BLOOM/mT5 — that check should happen before committing to this option) and more faithfully tests the theory's actual claim (reasoning *and* tool use, not one model doing both). Costs API spend, drops the fully-open constraint.
3. **Narrow scope**: run the test on one pair BLOOMZ already generates reliably (en-fr, 100% clean), treat low-resource generalization as future work.

Leaning toward option 2 — it resolves Phase 6's confound directly rather than continuing to search for a model that clears it. Option 3 is the fallback if preserving an open-only stack matters more than resolving the question now.

What changes in the paper draft

- **Sec. 5.3 (QE-in-the-loop)**: explicit caveat — doesn't work for 3/5 tested pairs; reframe as open problem.
- **Sec. 5.1 (triggering)**: same — cheap features show no signal; only crude implementations ruled out so far.
- **Sec. 1 (motivation)**: add the exposure-floor finding as a distinct failure mode from reasoning/tool insufficiency.
- **Core joint-necessity claim**, wherever stated: needs "confounded/untested cleanly" framing — not silence, not overclaiming from Phase 3.

Appendix: debugging record (compressed)

- **Aya-101 (Phase 3)**: T5 seq2seq, no chat template; flattened multi-turn context to a role-tagged transcript — fine for single-turn conditions, degenerated for the 4-step CoT chain.

- **Refine-step scoring bug (Phase 4)**: Qwen2.5-32B's refine responses were often full English explanations, entirely scored as "the translation" — a fake collapse from $0.71 \to 0.53$ COMET-KIWI. Fixed with an "output only the translation" instruction plus an extractor, validated against the original buggy data ($287/290$ cleaned correctly).
- **BLOOMZ/InkubaLM (Phase 5)**: 9 fixes — missing pre-download, an outdated KV-cache path in InkubaLM's custom model class, partial gated-repo caching, a context-window miscalculation, role-tag framing that made both models echo prompt boilerplate, a cue-removal that caused empty output, a misdiagnosed repetition-penalty fix, a cue-induced truncation bug, and finally a genuine correct-vs.-echo reliability problem that wasn't further fixable. An earlier full run that looked like a contradiction of Phase 3 was explicitly retracted once these bugs were understood.
- **ROOTS-table correction (Phase 6)**: a secondary characterization claimed Xhosa/Zulu were absent from BLOOM's pretraining data. The primary source (BLOOM paper, Table 1) shows both present with real, small byte counts. The claim that German has *zero* presence held up, though — and turned out to be the more consequential finding.